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1) Artificial intelligence (A.I) and Precision farming have been proposed as solutions to improve agricultural productivity in India. Critically analyze their feasibility and challenges in Indian agricultural landscape.

India's finance minister Nirmala Sitharaman has termed agriculture as the first and most crucial growth engine of Indian Economy.

As it is not only a traditional practice sector but it is also important for India's.

- 1) Food security
- 2) Rural economy
- 3) Self reliance,

Therefore budget 2025-26 has seen.

major policy shift and a new investment road map which will help agriculture to

- 1) Modern Agriculture
- 2) Increase productivity.

this will help in increasing productivity of crop yielding. by using Artificial intelligence

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and precision farming.

Precision farming :->

It is also known as precision agriculture, a technique that uses technology to increase crop yield and improve management by using sensors, Drones, satellite mapping, data analytics tools.

Feasibility Areas

- 1) land
- 2) labour
- 3) Electrification in rural areas.
or
Infrastructure.
- 4) Economic aspect / credit
- 5) Digital literacy.

challenges

Consolidation of land is still in progress.

skilling of labourers

Electrification is still under progress in many rural areas.

as most of our farmers base is lacking in initial capital investment.

lacking in necessary

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digital skill to operate
and interpret A.I based
system.

Other than this the challenges in following
precision agriculture is about

1) Institutes → less number of Agriculture
institute to ~~an~~ analyse data
and study agriculture map starts.

2) Ex:- ICAR, Agriculture universities, Krishi
Bhavan, IARI (Indian Agri Research Institute)

*) government policies and scheme to overcome
Challenges are :-

1) PM Dhan - Dhanya Krishi Yojana :-

- It is inspired from aspirational district program
to increase productivity in 100 low productivity
agricultural district.

- using A.I based precision farming,

- micro irrigation

NITI Aayog report $\Rightarrow$ 85% of Indian farmers have land holding less than 2 hectare. So for A.I and precision Agriculture land consolidation is a problem.

2) High yielding seeds $\Rightarrow$ [National Mission]

Introducing high grade quality seeds to improve yielding in production.

So, precision agriculture institutes will help in research development, in,

- 1) Climate resistant crop seeds.
- 2) Disease free seeds.
- 3) Nutrients level will be more to tackle nutritional aspects of India in fighting against hunger (SDG-1, 2).

3) Irrigation aspect $\Rightarrow$

As 60% of Indian irrigated land depend on traditional irrigation, pump set, tubewell, canals, so water wastage leads to low crop productivity.

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Micro irrigation system or drip irrigation system will ensure more productivity.

govt scheme: → "HAR KHET TAK PAANI" scheme.

↳ to fight drought

2) Rashtriya Krishi Vikas Yojana

- to focus more on effectively use of water through drip and sprinkler irrigation.

use of precision farming will help in improving agriculture productivity. with the upskilling of the farmers through Digital literacy programme, and providing initial economical support to shift from traditional to Artificial intelligence in agriculture practice will help in improving agricultural productivity and will ultimately help in achieving India Sustainable development goal of No poverty, zero hunger. and it will also place India among the top crop yielding with ~~top~~ quality countries.

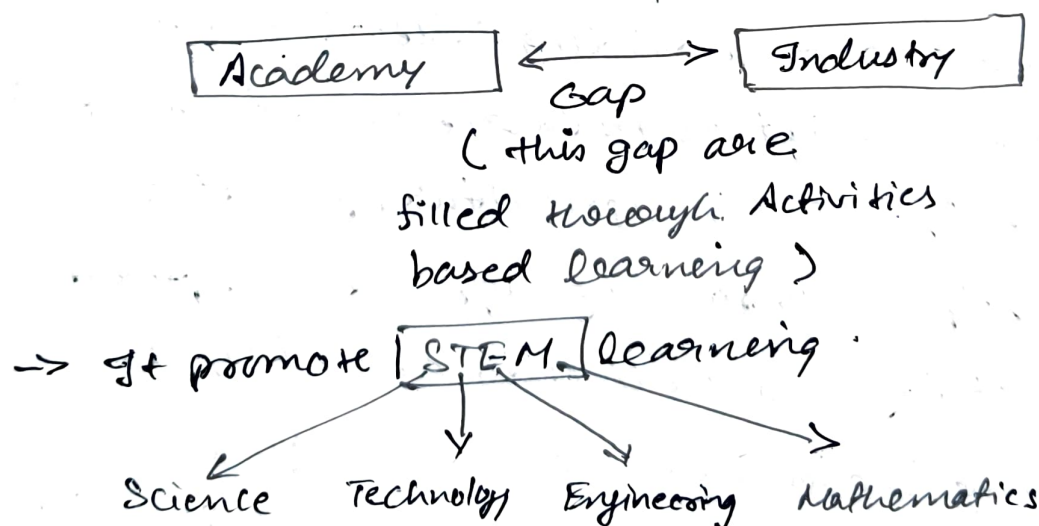
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Can Atal Tinkering Labs transformed STEM education in India? Analyse their potential impact on scientific temperament and innovation.

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Atal tinkering labs promote skilling or Activity based learning to fill the gap between theory and practical learning.

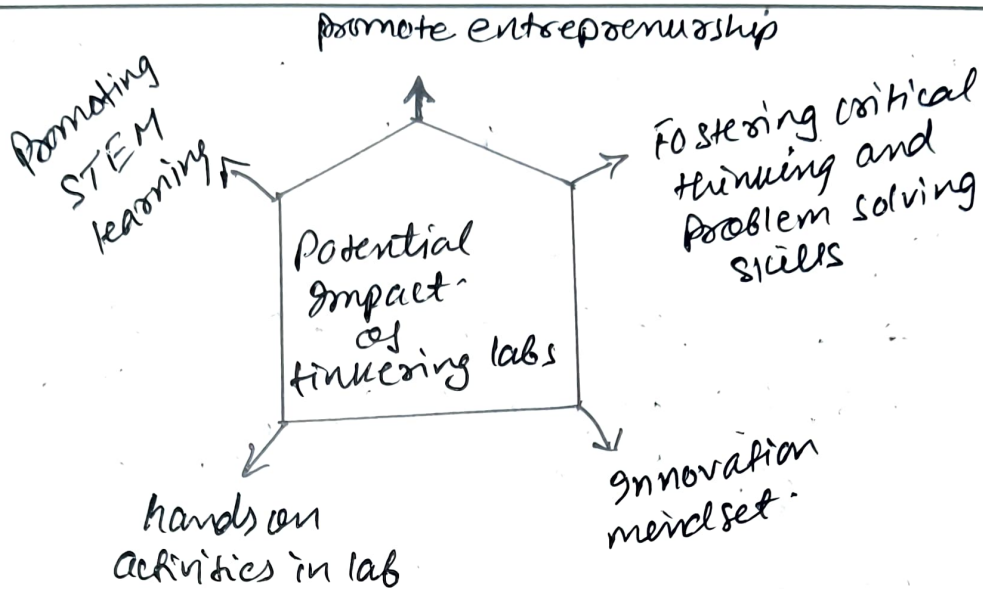
tinkering lab objective is to Nurture innovation and entrepreneurship in mindset of high school going children or students.



Due to this huge gap between Academic and Industry, the establishment of tinkering Lab will give industry ready student.

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As per Niti aayog data → tinkering lab improves critical and innovating thinking upto 30% among students.

Jean piaget → Book - "The science of student" and psychology of child.
↳ explains the use of ^{important} experiment learning.

Challenges :-

1) lacking in trained teachers - As per report 1.5 lakhs schools STEM teachers posts are vacant.

So, Number of teachers and their trainings has to be improved. through National teachers upskilling programs.

2) More thinking lab in school and at higher level has to developed.

3) language barrier :->

↳ A translation or translator of the subjects into mother language will also benefit in learning rural area students.

Benefits :->

- Will help India becoming global leader in STEM education.
- Will produce next generation innovators and technological leader.
- Will fill Academic and industry gap.
- Digital India dream and A.I driven technological future will get a boost.

thinking lab will help India to create its own innovation mission by fostering critical thinking and problem solving skills.

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How can National Centre of Excellence for Skilling bridge India's Skill gap? Critically analyze the structural and technological challenge they face.

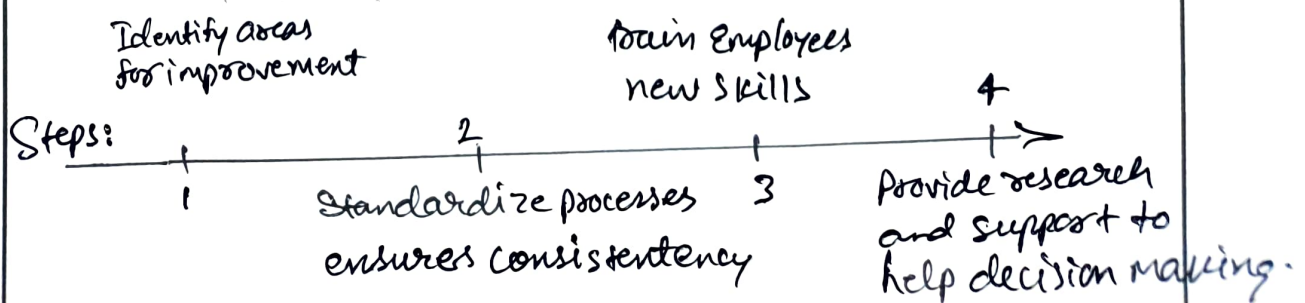
National Centre for Skilling is industry linked skilling program which help in upskilling or training or giving "hands on Activity" to workers.

COE is a set of activities provides leadership training and best practices for a specific area of focus.

COE helps to improve.

- ① Business procedure/processes.
- ② Performance
- ③ customer value.

How COE works :->



MIT Report 2023 → Every developed nations focus on research development and has well recognised institutes.

India is lagging in "Research and development" institutes and standard quality world recognised university.

Challenges :->

Structural

- 1) Lagging in Infrastructure
- 2) Credit / Loan facilities.
- 3) Innovation mindset.
- 4) Fear of investing due to small size industry at world level.
- 5) Lacking in government support in loss phase.

Technical

- 1) Digitally unaware
- 2) Lagging in use of A-I and Machine learning.
- 3) Cyber security.

→ As India's most of startup closes due to losses.

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to tackle this issues

- 1) Industry Academia collaboration is needed.
- 2) more institutes like IIT's and IIM and other Research institutes must be developed.
- 3) government investment should be priority in Research and development sector.
- 4) Reliance labs & in Artificial intelligence and Quantum Computing Research labs.

reforms :->

- ① establish international research collaboration fund.
- ② technological commercialisation
 - ↳ Research papers of IIT's and IIM's should be made to be utilised in Commercial market.
- ③ A.I for teachers upskilling and Student learning has to be promoted.
- ④ India if adopt it will become global leader in the world and COE programs will improve India's Industrial Ecosystem.

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Discuss the role of education and skill development in sustaining India's long term economic growth. How can Strategic investments in human capital bridge the gap between economic aspirations and realities?

Investment is one of the most important and biggest engine of GDP growth for any country. Investment is termed as the 3rd most important engine of growth.

In the ~~B~~ budget 2025-26 investment in people are sub-categorised into 12 initiatives. major are :-

- 1) Atal tinkering lab Expansion
- 2) Bhartiya Bhasha pushtak scheme
- 3) National Centre of Excellence for Skilling Scheme / program.
- 4) Expanding IIT infrastructure
- 5) Establishing centre of excellence in Artificial intelligence education.

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The role of education and skill development in sustaining india's long term economic growth is very crucial and important.



these sectors are prioritise by government and with the help of schemes and different programs, government is trying to tap the opportunity through:

1) Atal tinkering lab Mission :->

- to Foster and Nurture innovation and entrepreneurship mindset among high school students in india.

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- Prioritise hands on activity learning.
- fill gap between Academic ready student to industry ready student.
- Critical and Reasoning ability of a student will improve.
- entrepreneurship mindset will boost india's MSMEs ecosystem.

Niti Aayog 2023 $\Rightarrow$ Made a Report of contribution of tinkering lab in improving the mindset of student by 30%.

Sean piaget $\Rightarrow$ Book $\rightarrow$ The science of education and psychology of child
 $\hookrightarrow$ promoted the idea of Experiment learning

2) Bharatiya Bhasha pustak Schem $\Rightarrow$

- Making availability of Books on digital platforms.
- promoting Mother tongue learning for Cognitive development.

UNESCO report $\Rightarrow$ Mother tongue learning is 30% more efficient.

Book - Language and mind - Noam Chomsky
↳ promoted language learning.

So, Investment needs in :->

1) Translator's skilling - A.I based model
Can resolve these
issues.

As India's 90% of texts are still in English
language. So a faster and more efficient
translators by A.I must be included.

3) National centre of excellence program :-

- It is industry linked skilling program.
- providing training to workers to upskill
- Artificial intelligence and machine learning
has to be promoted.

Infrastructure :-

- 1) IIT's & IIM's Expansion
- 2) Training lab & Skilling lab Expansion
- 3) Teacher training
- 4) Cyber security & cloud computing.

As per, WEF report :->

8.5 crore jobs will be upcoming in digital and Automation sector till 2030. and If we didn't quench this opportunity to improve our infrastructure, educate and invest in education sector to give a platform to grow and nurture the human capital into a aspirational reality of true meaning of developed india @ 2047 will be a dream.